

Remediation 1 - Scripts / modularity

Use only for the failed competency

WHEN TO USE

Workflow depends on cell order, global state or one giant function.

TARGETED REPAIR

Rebuild a fresh 3-stage customer report as explicit load/validate/summary/write functions with clean process and missing-file test.

PASS SIGNAL

Pass when clean process and changed path work without hidden state.

Remediation 2 - pandas / files / output

Use only for the failed competency

WHEN TO USE

Types, file scope, lineage or output reconciliation is weak.

TARGETED REPAIR

Use a changed month folder with backup/drift plus malformed numeric. Produce file/schema/type controls and round-trip output reconciliation.

PASS SIGNAL

Pass when only approved rows flow and exceptions remain visible.

Remediation 3 - JSON / API / failure

Use only for the failed competency

WHEN TO USE

Nested grain, timeout/status handling or exception safety is weak.

TARGETED REPAIR

Flatten a changed event JSON and design an offline-first API wrapper. Inject timeout/HTTP/schema failures and prove final output is withheld when critical.

PASS SIGNAL

Pass when transport/data-quality failures cannot masquerade as success.

Remediation 4 - Logging / validation

Use only for the failed competency

WHEN TO USE

Logs exist but cannot explain trust decision.

TARGETED REPAIR

Create run-stage log plus structured validation JSON with observed/rule fields. Force a duplicate-key failure and show withheld output.

PASS SIGNAL

Pass when another analyst can reconstruct the run and reason for block.

Remediation 5 - Rerun / reproducibility

Use only for the failed competency

WHEN TO USE

Second run duplicates data or environment/input evidence is missing.

TARGETED REPAIR

Externalize config, record exact input hashes/versions/output hash, run twice, move project root, reproduce from clean process.

PASS SIGNAL

Pass when same approved inputs reproduce intended result without mutating raw data.